

Does L-BFGS-B Preserve the Relative Mean?

Read-only iteration audit of the two-chain Lennard-Jones convergence study

Continuum Limit Relaxation research audit

September 8, 2026

Audience. Continuum Limit Relaxation research project.

Decision. Does the current L-BFGS-B calculation preserve the free relative mean, and do the results justify applying Appendix B normalization before or during optimization rather than only after it?

Executive answer

No general relative-mean conservation was found. The parametrization preserves only

$$\bar{u}_1 + \bar{u}_2 = 0$$

at every evaluated state. Across all 36 traced solves, the largest observed gauge defect was 1.04×10^{-17} . The remaining scalar

$$\mu_k := \bar{u}_{1,k} = -\bar{u}_{2,k}$$

was free to change. The continuum-sampled track changed by at most 3.52×10^{-5} , and seed 3 changed by 6.57×10^{-4} to 7.86×10^{-4} . Other seeds changed materially more: up to 1.49×10^{-3} for seed 0, 1.30×10^{-3} for seed 1, 3.97×10^{-3} for seed 2, and 5.31×10^{-3} for seed 4. At $N = 640$, these are approximately 22%, 5.9%, 21%, 1.8%, and 25% of the respective initial mean magnitudes.

The evidence therefore rejects the claim that L-BFGS-B preserves relative mean as an algorithmic invariant. Seed 3 happens to retain it closely in this basin. The data do **not** justify inserting the discontinuous Appendix B map into every L-BFGS-B iteration. The production design should keep Appendix B as a post-solve representative selection. A one-time pre-normalization can be studied as a separate A/B experiment, but it would not make relative mean invariant during the solve.

1. Scope and method

- No repository source was edited. The existing user-owned docstring modification was left untouched.
- The live `relax` interpreter was used: Python 3.12.12, NumPy 2.4.2, SciPy 1.17.1.
- The saved 800-point continuum correction in `profiles.npz` supplied the continuum-sampled starts.
- Six starts were evaluated at each $N = 20, 40, 80, 160, 320, 640$: the continuum-sampled start and random-Fourier seeds 0 through 4.
- Every run used the production objective, analytic reduced gradient, cutoff $M = 80$, memory $m = 50$, `pgtol=5e-8`, `factr=1`, `maxiter=5000`, and `maxls=100`.
- An external callback observed every accepted major iterate x_k , reconstructed u_k with `build_mean_gauge`, and recorded both layer means. It did not observe rejected line-search trial evaluations. The [SciPy interface](#) defines the callback in this same after-each-iteration way.
- The continuum and seed-3 runs reproduced every saved normalized endpoint exactly: maximum array difference 0.0.

2. Production tracks: input, trajectory, and pre-normalization endpoint

Here `max excursion` is $\max_k |\mu_k - \mu_0|$ over accepted iterates. All means in this table are before Appendix B except the continuum cases where $d = 0$ and post-normalization is unchanged.

N	continuum μ_0	continuum μ_*	max excursion	seed-3 μ_0	seed-3 μ_*	seed-3 $\Delta\mu$	max excursion
20	1.39e-18	3.5170e-05	3.5172e-05	0.0441392	0.0433591	-0.0007801	0.0007845
40	-2.52e-18	4.7318e-06	4.7320e-06	0.0441392	0.0434820	-0.0006572	0.0006637
80	-2.30e-18	6.1398e-07	6.1398e-07	0.0441367	0.0434437	-0.0006929	0.0006938
160	-5.55e-18	7.8171e-08	7.8171e-08	0.0441196	0.0433808	-0.0007388	0.0007389
320	-5.55e-18	9.1204e-09	9.1744e-09	0.0441196	0.0433504	-0.0007692	0.0007693
640	0.00e+00	1.1447e-09	1.1815e-09	0.0441196	0.0433334	-0.0007863	0.0007864

The continuum track does not preserve zero exactly, but its displacement is very small and rapidly decreases with refinement. Seed 3 loses roughly 1.5% – 1.8% of its initial relative mean. Its apparent preservation is therefore approximate and basin-specific, not exact.

3. Seeds 0-4 as N doubles

3.1 Pre-Appendix-B endpoint mean

N	seed 0	seed 1	seed 2	seed 3	seed 4
20	0.006231	0.022844	0.015039	0.043359	-0.018193
40	0.007527	0.023340	0.015156	0.043482	-0.016910
80	0.007980	0.023384	0.015044	0.043444	-0.016364
160	0.008144	0.023423	0.014950	0.043381	-0.016122
320	0.008209	0.023433	0.014887	0.043350	-0.016011
640	0.008238	0.023436	0.014854	0.043333	-0.015956

Each seed approaches a seed-specific pre-normalization mean as N increases. This supports a shallow phase landscape with memory of the initialization; it does not show a conserved quantity.

3.2 Net change during L-BFGS-B

The entries are $\Delta\mu = \mu_* - \mu_0$.

N	seed 0	seed 1	seed 2	seed 3	seed 4
20	-0.000672	+0.000606	-0.003817	-0.000780	+0.003285
40	+0.000769	+0.001102	-0.003690	-0.000657	+0.004397
80	+0.001230	+0.001245	-0.003781	-0.000693	+0.004904
160	+0.001397	+0.001285	-0.003877	-0.000739	+0.005140
320	+0.001464	+0.001295	-0.003937	-0.000769	+0.005252
640	+0.001493	+0.001297	-0.003970	-0.000786	+0.005306

At the finest grid, seeds 0, 2, and 4 move by about one-fifth to one-quarter of their initial mean magnitude. Seed 3 is unusually stable among these starts.

3.3 Finest-grid trajectory ranges

start	μ_0	min accepted iterate	max accepted iterate	μ_*	iterations
continuum	0.000000000	-0.0000000005	0.0000000012	0.0000000011	313
seed 0	0.006744943	0.006744943	0.008238150	0.008238130	2327
seed 1	0.022138478	0.022138472	0.023436800	0.023435852	2156
seed 2	0.018824657	0.014854102	0.018824657	0.014854175	1747
seed 3	0.044119620	0.043333226	0.044119633	0.043333361	1393
seed 4	-0.021262087	-0.021262094	-0.015955540	-0.015955737	1384

The endpoint changes are not late numerical noise: the accepted-iterate ranges show systematic motion in the free mean direction, followed by settling.

4. What Appendix B changes after optimization

The table gives the first-layer mean after the exact Appendix B symmetry normalization. The second-layer mean is its negative.

N	seed 0	seed 1	seed 2	seed 3	seed 4
20	0.006231	0.022844	0.015039	0.043359	-0.018193
40	0.007527	0.023340	0.015156	-0.034088	-0.016910
80	0.007980	-0.015642	0.015044	0.004418	-0.016364
160	0.008144	0.003850	-0.004624	0.004233	0.003451
320	-0.001594	0.003829	-0.004717	0.004142	0.003594
640	-0.001572	-0.001089	0.000139	-0.000811	-0.001241

These jumps are changes of representative under an exact discrete symmetry, not jumps made by L-BFGS-B. Appendix B chooses $d \in \mathbb{Z}$ and a common translation c only after the optimizer returns, ensuring

$$\bar{u}_1 + \bar{u}_2 = 0, \quad |\bar{u}_i| \leq h/4.$$

5. Why L-BFGS-B does not conserve the relative mean

In the reduced coordinates, the objective is

$$F(a) = E(\text{build_mean_gauge}(a)).$$

The reconstruction enforces one linear condition, the sum-of-means gauge. Relative mean is another linear functional of a . A step p_k preserves it only if that functional annihilates the step. The current parameterization, objective callback, and L-BFGS-B call impose no such condition.

This matches the algorithmic specification. L-BFGS-B solves problems with componentwise bounds and otherwise updates the full parameter vector; the current call supplies no bounds. The [official SciPy documentation](#) lists only the supplied bounds as parameter constraints. The original Byrd-Lu-Nocedal-Zhu method constructs a line-search step and stores limited-memory correction pairs $s_k = x_{k+1} - x_k$ and $y_k = g_{k+1} - g_k$; it does not preserve an arbitrary unsupplied linear functional. See [A Limited-Memory Algorithm for Bound-Constrained Optimization](#), Sections 2-3.

6. Should Appendix B be applied before or during optimization?

Before the solve

A one-time Appendix B normalization of the initial state is mathematically admissible because it selects an energy-equivalent representative. It may be useful for a controlled diagnostic comparison. It would **not** cause L-BFGS-B to preserve relative mean afterward, because the relative-mean direction remains free.

No pre-normalized A/B sweep was run here: it would be a different experiment beyond the requested observation of the current algorithm.

During every iteration

Do not add this naively. The integer d changes through nearest-integer rounding, so the canonical representative map is discontinuous at cell boundaries and includes a layer-index roll. L-BFGS-B's stored correction pairs describe differences in one coordinate chart. Replacing an iterate by a differently rolled/shifted representative without consistently transporting or restarting the quasi-Newton memory can make those pairs describe the wrong coordinate differences. This is an inference from the Appendix B map and the documented L-BFGS-B correction-pair mechanism, not a directly tested failure in this repository.

A quotient-aware optimizer could canonicalize representatives safely only with an explicit design for gradient transport, history transport or restart, boundary crossings, and equivalence tests. That is a substantive algorithmic change, not a simple validation check.

Recommendation

1. Keep the exact sum-of-means gauge inside every objective/gradient evaluation.
2. Keep Appendix B normalization after convergence for the production calculation.
3. Add relative-mean tracing as a diagnostic when studying phase behavior; report μ_0 , μ_* , maximum excursion, d , and the post-normalization mean.
4. If desired, separately test one-time pre-normalization versus the current starts and compare energy, stationarity, iteration count, and symmetry-aligned endpoints.
5. Do not constrain both layer means to zero: that would remove the physical relative-mean direction rather than merely select an exact symmetry representative.

7. Numerical quality and limitations

- All 36 runs returned `warnflag=0`.
- The largest endpoint Euler-Lagrange residual divided by h was 9.635×10^{-3} , below the current 10^{-2} acceptance threshold.
- Some solves stopped by relative function reduction with reduced-gradient infinity norms above `pgtol`; the largest was 1.445×10^{-6} . Results should therefore be read under the current endpoint policy, not as proof that every run met `pgtol`.
- Seeds 0-4 are a small deterministic ensemble, not representative random-start statistics.
- The callback records accepted major iterates, not rejected line-search trials.
- The audit diagnoses the existing algorithm. It does not test a modified quotient-aware or per-iteration-normalized optimizer, prove global minimality, or verify the manuscript's stability-gap hypothesis.

Sources and evidence ledger

1. The live study script, `profiles.npz`, and `convergence_summary.csv`, accessed September 8, 2026. Used for the coordinate map, starts, solver settings, continuum reference, and reproduction checks.
2. *Continuum Limit of a One-Dimensional Atomistic Model for Moire Relaxation*, local manuscript `Hao_OralExam_Continuum_Limit.pdf`, July 21, 2026. Used for the discrete symmetry and Appendix B result.
3. [SciPy `fmin\_l\_bfgs\_b` documentation](#), accessed September 8, 2026. Used for callback timing, bounds, memory, and stopping claims.
4. Byrd, Lu, Nocedal, and Zhu, *A Limited-Memory Algorithm for Bound-Constrained Optimization*, *SIAM Journal on Scientific Computing* 16(5), 1995, pp. 1190-1208. Used for the line-search and correction-pair mechanism.

Stopping rationale. The callback experiment answers the conservation question, alternative seeds agree, saved outputs reproduce exactly, and primary sources explain the optimizer mechanism.